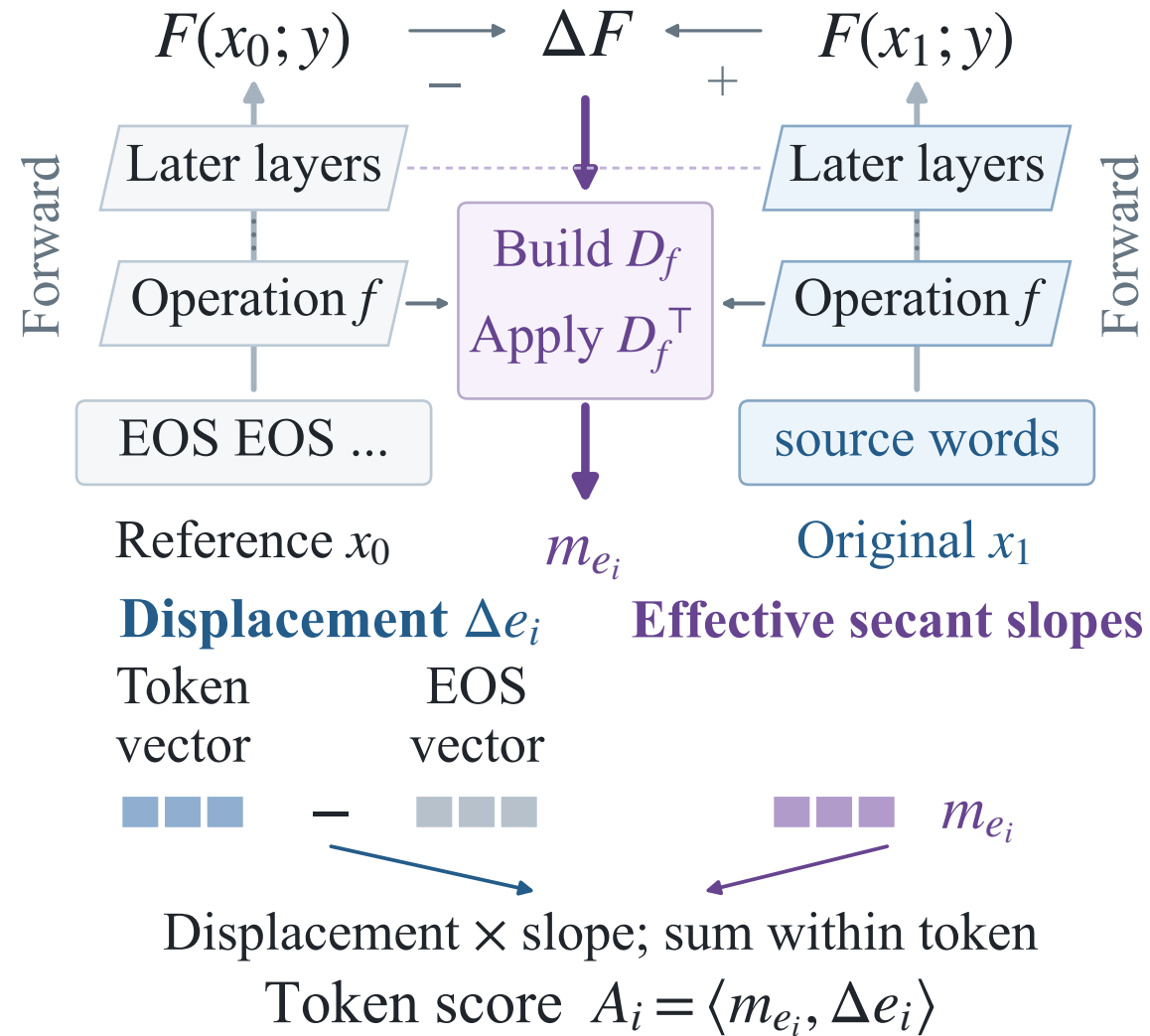


(a) Overall method

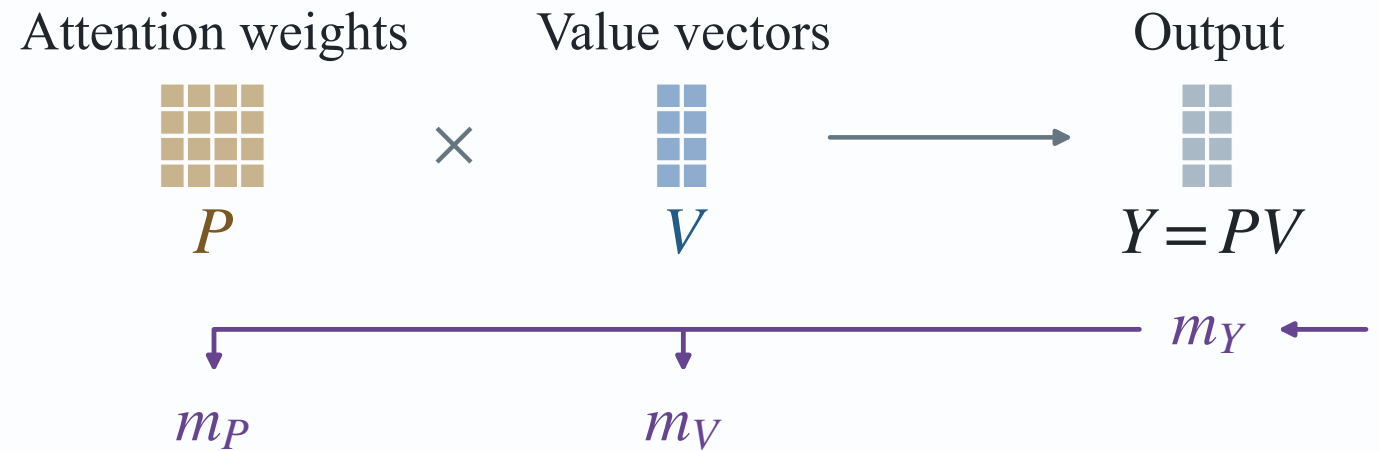
Two forward runs with fixed response y



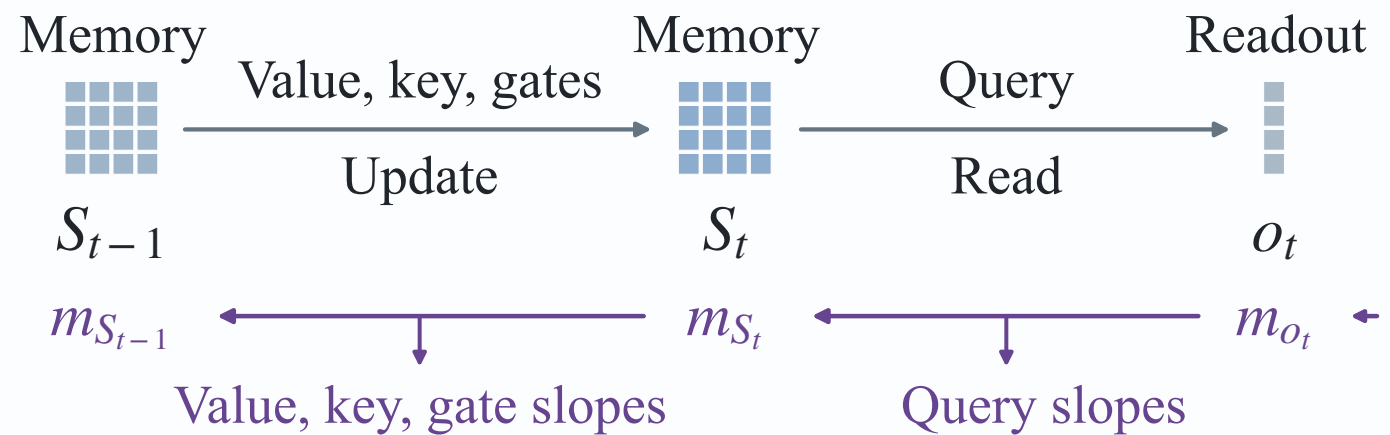
Reverse-pass examples

→ Forward ← Backward $D_f^\top$

(b) Attention product



(c) Gated DeltaNet



(d) Choose the playwright, not the composer.

CONTEXT EXCERPTS

... William Shakespeare's play

... The score is by William Walton.

REQUESTED ROLE (EXCERPT)

... the author of the play...



Name-span total (nats)

+1.56

-1.13

Token contribution (nats)

≤ -2.5 0 $\geq +2.5$

Shown spans underlined

MoreHopQA example 1

Answer

William
Shakespeare

$$\sum_i A_i = \Delta F$$

Across all input sources